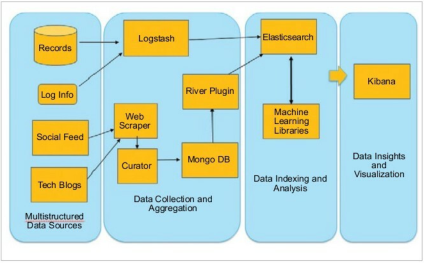

Figure 1: Block diagram of an open source data science stack

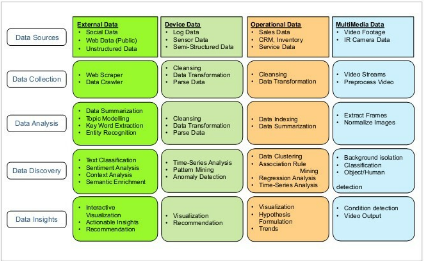

Figure 2: Block diagram of various stages of data

creating dashboards. MongoDB is a document database that can handle unstructured data. NLTK and Scikit are software libraries performing natural language parsing, tokenisation, automatic keyword extraction and machine learning. Figure 1 illustrates a data science stack with open source components.

Data stages

We consider both structured and unstructured data resources. The in-house data has structured sources of information. They are from databases comprising inventory, sales information pertaining to the stocks on hand, item numbers, average costs, availability, location, etc. The data from blogs is unstructured and provides us insights into purchasing behaviour, sentiments about the products, their quality and services offered by the enterprise. A typical data flow using multi-structured data with open source components is shown in Figure 2. It illustrates the various input resources for collecting data in an enterprise to get insights for intelligent product allocation, to improve the customer experience, to develop and execute marketing campaigns and to initiate product enhancements.

The analysis, discovery and insights for the various data sources are given in the diagram (see Figure 2). These stages are realised using the open source stack shown in Figure 1. The enterprise, like a retail store, can get insights by amalgamating data from external sources such as the Internet for unstructured data – blogs, tech sources, social media – and from internal sources comprising log data from machines, databases, devices and applications, as well as structured data from databases.

Data Science Methodology

The data science methodology is described in the following paragraphs.

Data extraction: The unstructured data from blogs and tech blogs are extracted using scrapers in Python. The hyperlinks for these blogs are either pre-determined or those which can be intelligently obtained through a crawler, which is a script to browse the Web automatically. Scrapy is a tool in Python used for scraping data for instance, let's consider a scenario in the retail industry, where we can fetch information on leather bags based on different brands and models. These can be further segmented based on a number of features. A feature set could include comments based on the colour and shape of the bag, design, and type of the bag (such as whether it's for women, men or children). The bag could be of use in offices, homes or parties. Scrapy is a tool in Python used for scraping data.

Data curatorship and archival: The extracted data can be curated to remove special characters, disclaimers, and filter redundant messages present in the original data. Intermediate data could be stored in temporary files. The curated data from temporary storage is archived in MongoDB through the Python interface. The data from Mongo DB is pushed into Elasticsearch through a Mongo River plugin whenever there are new data changes.

Data model: The unstructured messages in the MongoDB fields are analyzed for sentiments using machine learning libraries in NLTK, R.

A supervised training model is created from the curated data using the available machine learning algorithms such as Naïve Bayes and Support Vector Machine (SVM). The accuracy for the classifier model is computed, and can be